

EXP NO 14: Implement House price Prediction using appropriate machine learning algorithm

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AIM

To implement a **House Price Prediction system** using a suitable **Machine Learning regression algorithm** and predict house prices based on various features.

PROCEDURE

1. **Collect and load** the house price dataset using Pandas.
2. **Preprocess the data** by handling missing values and selecting relevant features.
3. **Split the dataset** into training and testing sets.
4. **Train a Linear Regression model** using the training data and predict house prices.
5. **Evaluate the model** using MAE, MSE, RMSE, and R^2 score, and display the prediction results.

CODE:

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, r2_score

# Dataset

data = {
    'Area': [1200, 1500, 1800, 2000, 2500, 3000, 3500, 4000],
    'Bedrooms': [2, 3, 3, 4, 4, 5, 5, 6],
    'Bathrooms': [1, 2, 2, 3, 3, 4, 4, 5],
    ...# Predict a new house price
    new_house = [[2200, 4, 3]]
    prediction = model.predict(new_house)
```

```
print("Predicted Price for New House:", round(prediction[0], 2), "Lakhs")
```

OUTPUT:

```
Actual Prices: [40 90]
Predicted Prices: [42.56 90.64]
Mean Absolute Error: 1.6
R2 Score: 0.99
Predicted Price for New House: 65.26 Lakhs
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: U
warnings.warn(
```

RESULT:

The House Price Prediction system was successfully implemented using the Linear Regression algorithm. The model was trained using house features such as area, bedrooms, and bathrooms, and successfully predicted house prices. The model performance was evaluated using MAE and R^2 score.

